

This is the non-calculator section!

Period _____

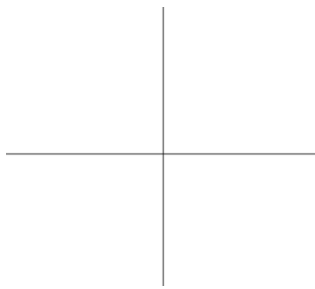
For 1 and 2, solve for x , leaving answers in exact form with powers and logs. [3 pts. each]

1. $2 \ln \ln (\sqrt{x}) - \ln \ln (1 - x) = 2$

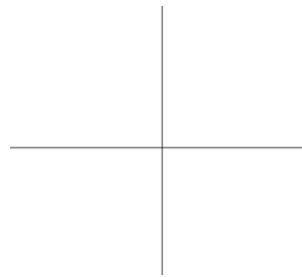
2. $\frac{10-3^x}{4} + 5 = -20$

Draw a rough sketch of the following 2 power functions. We are looking for shapes and domain and ranges so your graphs don't have to be super exact. [3 pts. each]

3. $f(x) = (x)^{-\frac{5}{9}}$



4. $f(x) = -(x + 4)^{\frac{4}{11}}$

5. Write an equation of a logistic curve in the form $f(x) = \frac{M}{1 + Be^{-kx}}$ that has a point of inflection at (40, 2000). [2 pts]

6. Given the following logistic function: $f(x) = \frac{60}{1+5e^{-0.2x}}$

a) Find the coordinates of the y-intercept and determine the carrying capacity. [2]

y intercept _____ Carrying capacity _____

Find the inverse function $f^{-1}(x)$ by switching x and y and solving for y. [4 pts]

b) Write equations of all asymptotes for $f^{-1}(x)$. [1 pt]

Continue on to the calculator section. Before taking out your calculator you must turn in this non-calculator section.